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(54) **MAGNETIC PIGMENT FLAKE, OPTICALLY VARIABLE INK, AND ANTI-FALSIFICATION ARTICLE**

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Feb. 24, 2021 (CN) 202110210021.6

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See application file for complete search history.

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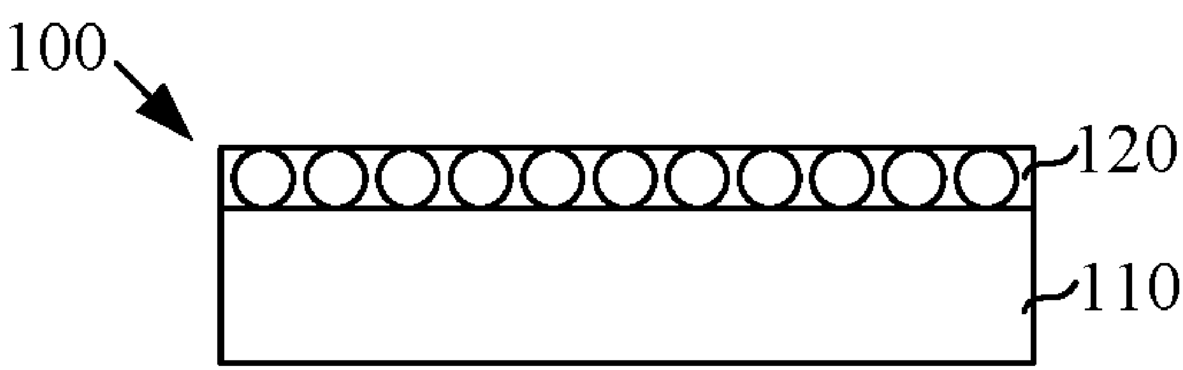
Primary Examiner — Alexandre F Ferre

(57) **ABSTRACT**

A magnetic pigment flake includes a filtering film layer, with magnetic or magnetizable material, and a metal nanoparticles layer, formed on a surface of the filtering film layer. The metal nanoparticles layer is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity. An optically variable ink includes an ink body and the above-mentioned magnetic pigment flakes. An anti-falsification article includes an article body and the above-mentioned optically variable ink. The magnetic pigment flake of the optically variable ink is magnetically oriented, such that a bright and dark areas are generated with a viewing angel changing under an irradiation of visible light below the predetermined intensity. Under an irradiation of visible light exceeding the predetermined intensity, light with a color different from that of the bright area is generated on a corresponding position of the dark area.

19 Claims, 5 Drawing Sheets



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C09C 1/00	(2006.01)
C09D 11/037	(2014.01)
C09D 11/50	(2014.01)
G02B 5/02	(2006.01)
G02B 5/22	(2006.01)
G02B 5/28	(2006.01)

(52) **U.S. Cl.**
CPC ***C09C 1/0015*** (2013.01); ***C09D 11/037*** (2013.01); ***C09D 11/50*** (2013.01); ***G02B 5/008*** (2013.01); ***G02B 5/0284*** (2013.01); ***G02B 5/22*** (2013.01); ***G02B 5/28*** (2013.01); *C01P 2004/20* (2013.01); *C01P 2004/80* (2013.01); *C01P 2006/42* (2013.01); *G02B 2207/101* (2013.01)

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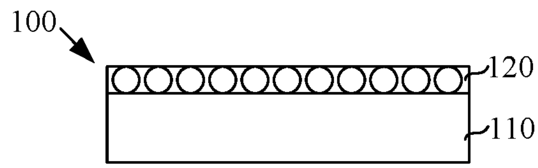

FIG. 1

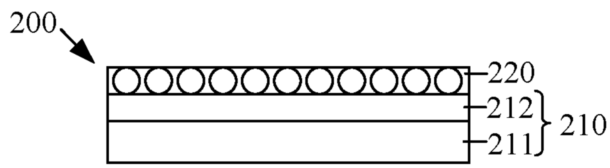

FIG. 2

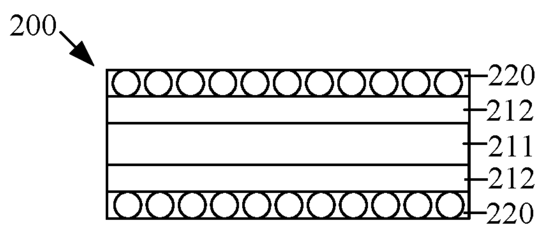

FIG. 3

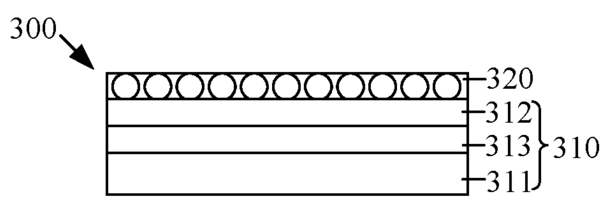

FIG. 4

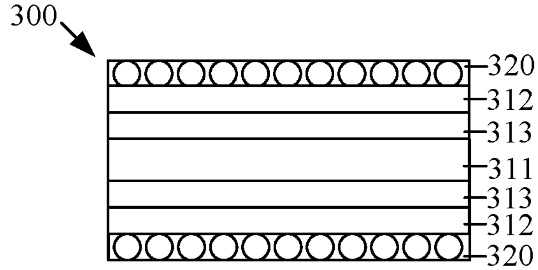

FIG. 5

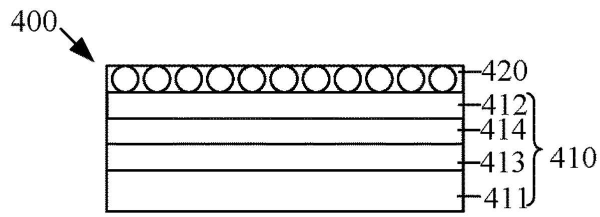

FIG. 6

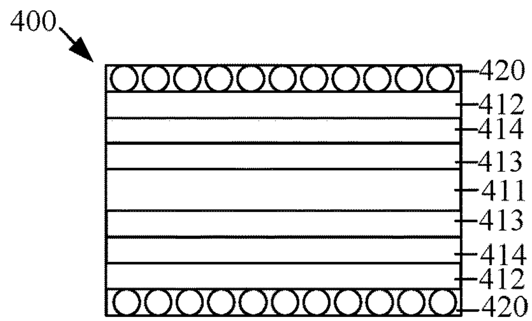

FIG. 7

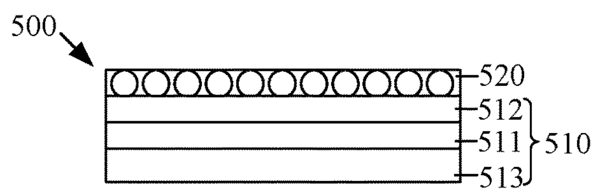

FIG. 8

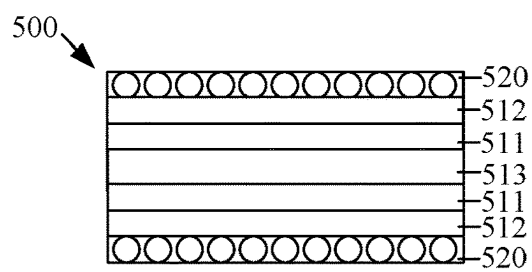

FIG. 9

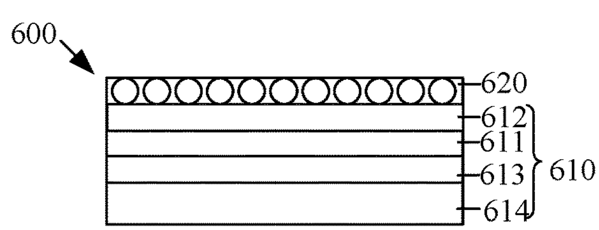

FIG. 10

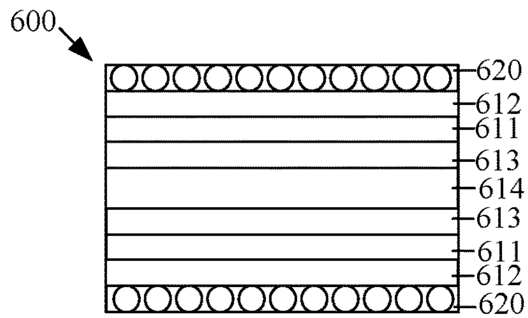

FIG. 11

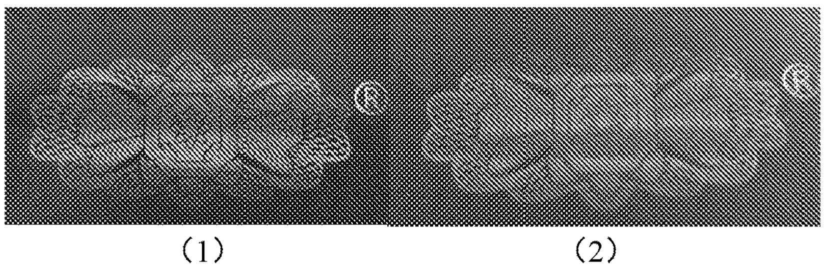

FIG. 12

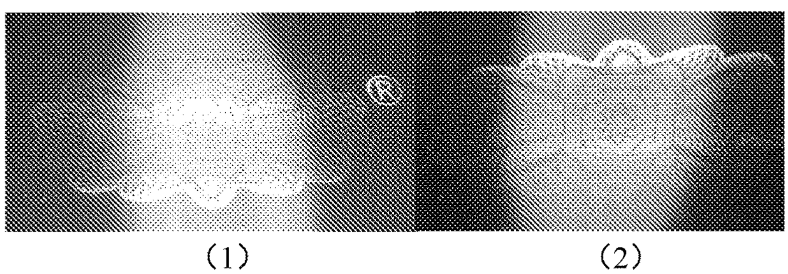

FIG. 13

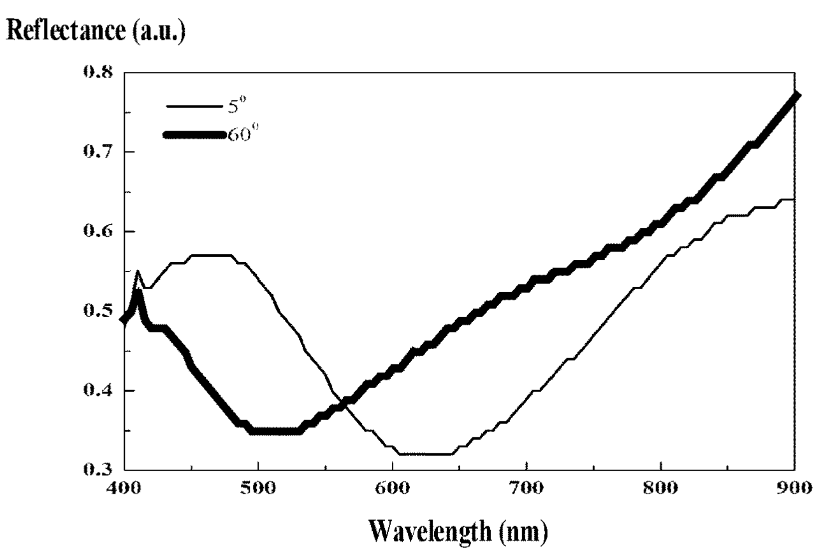

FIG. 14

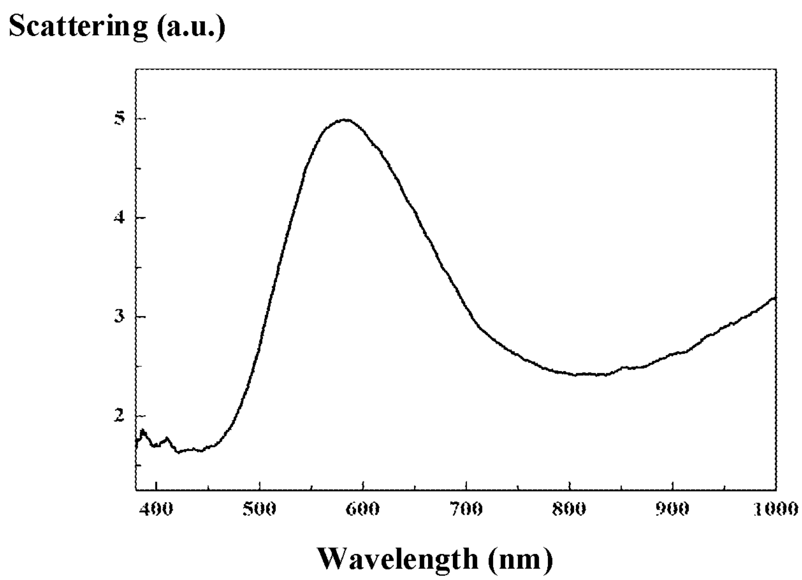
Scattering (a.u.)
5
4
3
2
400
500
600
700
800
900
1000
Wavelength (nm)

FIG. 15

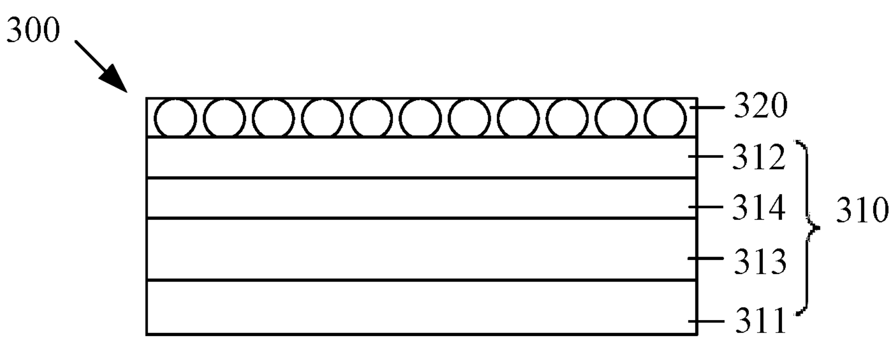

FIG. 16

MAGNETIC PIGMENT FLAKE, OPTICALLY VARIABLE INK, AND ANTI-FALSIFICATION ARTICLE

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of International Patent Application No. PCT/CN2021/103887, filed Jun. 30, 2021, which claims priority to Chinese Patent Application No. 202110210021.6, filed Feb. 24, 2021, the entire disclosures of which are incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to the field of a magnetic orientation, and in particular to a magnetic pigment flake, an optically variable ink, and an anti-falsification article.

BACKGROUND

Based on a principle of an optical thin film interference, a color of an optically variable film changes at different viewing angles. Due to a unique flip-flap property of the optically variable film, and a bright color changes obviously, the optically variable film cannot be scanned and copied, and is easily recognized by the public. The optically variable film has been widely used in an anti-falsification field of banknotes, securities and cigarette packets. With an optically variable anti-falsification technology, a circulation of counterfeit articles in the market is effectively reduced, and development of a falsification technology is curbed, thus it played an important role in maintaining the stability of market economy and information security. However, structures, action mechanisms and technological processes of magnetic optically variable pigments have been reported by a large number of patents and academic papers. At the same time, the magnetic optically variable pigment has also been applied in the decoration market, such that a credibility and anti-falsification performance as a high-security document protection are gradually decreasing.

SUMMARY OF THE DISCLOSURE

According to a first aspect of the present disclosure, a magnetic pigment flake is provided and includes: a filtering film layer, with a magnetic or magnetizable material; and a metal nanoparticles layer, formed on a surface of the filtering film layer. The metal nanoparticles layer is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity.

According to a second aspect of the present disclosure, an optically variable ink is provided and includes: an ink body; and the magnetic pigment flakes according to any one of above embodiments. The magnetic pigment flakes are doped in the ink body.

According to a third aspect of the present disclosure, an anti-falsification article is provided and includes: an article body, and the optically variable ink according to any one of above embodiments. The optically variable ink is coated on the article body. The magnetic pigment flake of the optically variable ink is magnetically oriented, such that a bright and dark areas are generated with a viewing angel changing under an irradiation of visible light below the predetermined intensity. Under an irradiation of visible light exceeding the predetermined intensity, light with a color different from that of the bright area is generated on a corresponding position of the dark area.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1** is a structural view of a magnetic pigment flake according to an embodiment of the present disclosure.

FIG. **2** is a structural view of the magnetic pigment flake according to another embodiment of the present disclosure.

FIG. **3** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **4** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **5** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **6** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **7** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **8** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **9** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **10** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **11** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

FIG. **12** is a schematic view of an anti-falsification pattern made of an optically variable ink of a magnetic pigment flake in a comparative example of the present disclosure.

FIG. **13** is a schematic view of an anti-falsification pattern made of an optically variable ink of the magnetic pigment flake in a first example of the present disclosure.

FIG. **14** is a schematic view of a reflectance-wavelength spectrum of the anti-falsification pattern made of the optically variable ink of the magnetic pigment flake in the first example t of the present disclosure.

FIG. **15** is a schematic diagram of a scattering spectrum of the magnetic pigment flake after scratching in the first example of the present disclosure.

FIG. **16** is a structural view of the magnetic pigment flake according to yet another embodiment of the present disclosure.

DETAILED DESCRIPTION

The disclosure will now be described in detail with reference to the accompanying drawings and examples. Apparently, the described embodiments are only a part of the embodiments of the present disclosure, not all of the embodiments. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present invention without creative efforts shall fall within the protection scope of the present invention.

It should be noted that if there are directional indications (such as up, down, left, right, front, back, etc.) involved in the embodiments of the present disclosure, the directional indications are only used to explain a relative position relationship and movement between components in a specific posture (as shown in the attached figures). If the specific posture changes, the directional indications are also changed accordingly.

In addition, if there is a description of "first", "second" and the like in the embodiment of the present disclosure, the description of "first", "second" and the like is only used herein for purposes of description, and are not intended to indicate or imply relative importance or significance or to imply the number of indicated technical features. Thus, the feature defined with "first", "second", and the like may include one or more of such a feature. Besides, technical solutions between various embodiments may be combined with each other, but it must be based on the realization by one skilled in the art. When a combination of the technical solutions is contradictory or cannot be achieved, it should be considered that the combination of the technical solutions does not exist and is not are included in the scope of protection of the present disclosure.

According to a first aspect of the present disclosure, a magnetic pigment flake is provided and includes: a filtering film layer, with a magnetic or magnetizable material; and a metal nanoparticles layer, formed on a surface of the filtering film layer. The metal nanoparticles layer is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity.

In an embodiment, a color of the scattered light generated by the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light exceeding the predetermined intensity is different from a color of light generated by the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light below the predetermined intensity.

In an embodiment, the filtering film layer comprises a magnetic core layer and a first medium layer; the first medium layer is stacked on at least one of main surfaces of the magnetic core layer; the metal nanoparticles layer is arranged on a side of the first medium layer away from the magnetic core layer.

In an embodiment, the filtering film layer further comprises a second medium layer; the second medium layer is arranged between the first medium layer and the magnetic core layer; a refraction index of the second medium layer is less than a refraction index of the first medium layer.

In an embodiment, the refraction index of the second medium layer is less than or equal to 1.65. The second medium layer is selected from at least one of silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, calcium fluoride, lithium fluoride, polystyrene, polyethylene, polymethyl methacrylate, polyamide imide, polyperfluoroethylene propylene, tetrafluoroethylene, trifluorochloroethylene, cellulose propionate, cellulose acetate, cellulose acetate butyrate, methylpentene polymer, homoformaldehyde, acrylic resin, cellulose nitrate, ethyl cellulose, polypropylene, polysulfone, poly-ethersulfone, mica, heterogeneous isomorphic polymer, polybutene, ionic cross-linked polymer, acrylic copolymer, thermoplastic, styrene butadiene, polyvinyl chloride, urea formaldehyde, styrene acrylonitrile, and polycarbonate.

In an embodiment, the filtering film layer further comprises an absorbing layer and a second medium layer; the absorbing layer and the second medium layer are arranged between the magnetic core layer and the first medium layer; the absorbing layer is closer to the first medium layer than the second medium layer.

In an embodiment, a thickness of the absorbing layer is less than 30 nm. Materials of the absorbing layer are selected from at least one of titanium, aluminum, chromium, nickel, palladium, titanium, vanadium, cobalt, iron, carbon, tin, tungsten, molybdenum, rhodium, and niobium, or selected from an alloy of at least one thereof, or selected from silicon carbide.

In an embodiment, the filtering film layer further comprises a second medium layer. The number of the magnetic core layer is at least two. The second medium layer is arranged between two adjacent magnetic core layers. The first medium layer is arranged on a main surface of the outermost magnetic core layer away from the second medium layer.

In an embodiment, the filtering film layer further comprises a second medium layer and a reflective layer. The number of the magnetic core layer is at least two. The second medium layer and the reflective layer are arranged between two adjacent magnetic core layers. The reflective layer is separated from each of the two adjacent magnetic core layers by the second medium layer. The first medium layer is arranged on a main surface of the outermost magnetic core layer away from the second medium layer.

In an embodiment, materials of the reflective layer are selected from at least one of aluminum, silver, gold, copper, platinum, tin, titanium, palladium, rhodium, niobium, chromium, and alloys thereof. A physical thickness of the reflective layer is in a range from 2 nm to 500 nm.

In an embodiment, an interference cavity is formed by the magnetic core layer and the first medium layer.

In an embodiment, a physical thickness of the magnetic core layer is less than 30 nm. The second medium layer is a dielectric stack with alternating high and low refractive index film layers, or a dielectric film layer. Materials of the second medium layer are selected from at least one of silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, and calcium fluoride.

In an embodiment, the metal nanoparticles of the metal nanoparticles layer are spaced apart from each other, and a gap of two adjacent metal nanoparticles is in a range from 2 nm to 1 mm.

In an embodiment, a particle size of the metal nanoparticles is in a range from 2 nm to 1 μm. Materials of the metal nanoparticles are selected from at least one of aluminum, silver, gold, copper, platinum, ruthenium, palladium, rhodium, cobalt, iron, nickel, lead, osmium, iridium, and alloys thereof. A shape of the metal nanoparticles is sphere, hemisphere, ellipsoid, cube, cuboid, octahedron, dodecahedron, hexahedron, bar, star, cone, triangle, or cylinder.

In an embodiment, a refraction index of the first medium layer is greater than 1.65.

In an embodiment, the first medium layer is selected from at least one of lanthanum titanate, titanium pentoxide, niobium pentoxide, zinc sulfide, zinc oxide, zirconia, titanium dioxide, carbon, indium oxide, indium tin oxide, tantalum pentoxide, cerium oxide, yttrium oxide, europium oxide, iron oxide, ferroferric oxide, hafnium nitride, hafnium carbide, hafnium oxide, lanthanum oxide, magnesium oxide, neodymium oxide, praseodymium oxide, samarium oxide, antimony trioxide, silicon carbide, silicon nitride, silicon monoxide, selenium trioxide, tin oxide, and tungsten trioxide. The magnetic core layer is made of at least one of iron, cobalt, nickel, gadolinium, terbium, dysprosium and erbium, oxides and alloys thereof.

In an embodiment, the metal nanoparticles layer is arranged on main surfaces of a front and rear sides of the filtering film layer, such that the magnetic pigment flake is arranged with a symmetrical structure centering on the filter film layer.

According to a second aspect of the present disclosure, an optically variable ink is provided and includes: an ink body; and the magnetic pigment flakes according to any one of above embodiments. The magnetic pigment flakes are doped in the ink body.

According to a third aspect of the present disclosure, an anti-falsification article is provided and includes: an article body, and the optically variable ink according to any one of above embodiments. The optically variable ink is coated on the article body. The magnetic pigment flake of the optically variable ink is magnetically oriented, such that a bright and dark areas are generated with a viewing angel changing under an irradiation of visible light below the predetermined intensity. Under an irradiation of the visible light exceeding the predetermined intensity, light with a color different from that of the bright area is generated on a corresponding position of the dark area.

In an embodiment, the anti-falsification article is configured to generate scattered light enhanced by a the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity, such that a color generated by the dark area of the anti-falsification article under an irradiation of the light exceeding the predetermined intensity is different from a color generated by the dark area of the anti-falsification article under an irradiation of the light below the predetermined intensity.

As shown in FIG. **1**, FIG. **1** is a structural view of a magnetic pigment flake according to a first embodiment of the present disclosure. The magnetic pigment flake **100** includes a filtering film layer **110** having a magnetic or magnetizable material and a metal nanoparticles layer **120**.

The metal nanoparticles layer **120** is formed on a surface of the filtering film layer **110** and configured to generate a local surface plasmon resonance (LSPR) under an irradiation of light exceeding a predetermined intensity, and light scatter-enhancing property with a wavelength selectivity for incident light is generated. The magnetic or magnetizable material of the filtering film layer **110** is directionally arranged in a magnetic field, such that an anti-falsification pattern formed by the filtering film layer **110** generates an obvious shading area, that is, a bright and dark areas. At the same time, cooperating with the light scattering enhancement property of the LSPR of the metal nanoparticles layer **120**, a color of the dark area of the anti-falsification pattern formed by the magnetic pigment flake **100** of the present disclosure and irradiated under ambient light is different from a color of the dark area of the anti-falsification pattern irradiated under the visible light exceeding the predetermined intensity. In some embodiments, the ambient light may be visible light below the predetermined intensity. In this case, the anti-falsification pattern formed by the magnetic pigment flake **100** of the present disclosure includes two colors changes in a front view and a side view based on a flip-flap effect of the magnetic or magnetizable material. Further, a third color (that is, a hidden color) is generated by the dark area of the anti-falsification pattern under the irradiation of the light exceeding the predetermined intensity, thus an anti-falsification and imitation difficulty of the anti-falsification pattern formed by the magnetic pigment flake **100** may be effectively improved.

In some embodiments of the present disclosure, the color development of the metal nanoparticles under high-intensity white light is the color of the scattered light enhanced by the local surface plasmon resonance, which is different from a color development by means of the local surface plasmon resonance of nanoparticles. Furthermore, only when a film layer (an outermost film layer of an interference film stack) directly contacted with the metal nanoparticles is a film layer with a high refraction index, and the high refraction index is greater than 1.65, a discoloration effect of the high-intensity white light can be achieved.

When a film layer of SiO_2 is arranged with a layer of Ag NPs, or a surface of a film stack of $MgF_2/AL_2O_3/MgF_2$ is arranged with the layer of the Ag NPs, a coloration is generated by a structure as described above due to wavelength-selective light absorption of the local surface plasmon resonance of the Ag NPs. However, the discoloration is not generated under the high-intensity white light. In addition, when a surface of a single film layer of ZrO_2 is arranged with the layer of the Ag NPs, the discoloration is also not generated by such structure under the high-intensity white light. In contrast to light absorption of the LSPR used in most of plasmon display devices, some embodiments of the present disclosure use the light scattering enhancement property of the LSPR.

The realization of a scattering color visible to naked eyes has strict requirements on a structure of a F-P film stack. Likewise, there are certain requirements for the ambient light. A bright environment is needed to observe a bright color of a traditional optically variable pigment of an interference type. In some embodiments of the present disclosure, a photochromic effect is generated by an optically variable pigment of a scattered type when the optically variable pigment of the scattered type is irradiated by the high-intensity white light in a dark environment. When an ambient illumination is lower than 300 lx and an illumination of predetermined light source is greater than 3000 lx (preferably, the illumination is greater than 5000 lx), it is possible to generate a better discoloration effect under the high-intensity white light. However, in a sunny outdoor environment, if the ambient illumination is greater than 3000 lx, the discoloration effect under the high-intensity white light is not generated by a sample. In some embodiments of the present disclosure, the optically variable pigment is a pigment generating a color change when the pigment is irradiated by the high-intensity white light in the dark environment, which is different traditional optically variable pigment. The color is changed from a scattered color under the ambient light to a scattered color under the high-intensity white light. In some embodiments, a color temperature of the light source is in a range from 4000K to 8000K.

In some embodiments, a reflection color of light generated by the metal nanoparticles layer **120** irradiated under the ambient light is different from a scattered color of light generated by the metal nanoparticles layer **120** irradiated under the visible light exceeding the predetermined intensity. When the metal nanoparticles layer **120** is irradiated under the visible light exceeding the predetermined intensity, the light passing through the metal nanoparticles layer **120** enters into the filter film layer **110**. The filter film layer **110** reflects and interferes the light, such that the light is reflected to the metal nanoparticles layer **120** again, such that the scattered light is further enhanced and modulated. In this case, the metal nanoparticles layer **120** disposed on the surface of the filtering film layer **110** has the light scatter-enhancing property with the wavelength selectivity for the incident light, thus the color of the pigment is changed from reflection color under the ambient light to the scattered color under light of the predetermined intensity.

In some embodiments, a wavelength range of irradiation light is necessary to be overlapped with the wavelength range of the SPR absorption peak of the metal nanoparticles, such that the LSPR is generated by the metal nanoparticles layer **120** under the irradiation light. In some embodiments, the irradiation light may be the visible light, and a wavelength range of the visible light is wide. The LSPR is generated the scattered light enhanced by the metal nanoparticles layer **120**, when the visible light with a sufficient intensity is in cooperation with the filter film layer **110**.

In some embodiments, materials of the metal nanoparticles layer **120** may be selected from at least one of aluminum, silver, gold, copper, platinum, ruthenium, palladium, rhodium, cobalt, iron, nickel, lead, osmium, iridium, and alloys thereof.

In some embodiments, the SPR absorption peak of the metal nanoparticles in the metal nanoparticles layer **120** is related to a shape and size of the metal nanoparticles. In this case, by changing the shape and/or size of the metal nanoparticles, the anti-falsification pattern changes based on the color change generated by the LSPR, thereby further improving the anti-falsification and imitation difficulty of the anti-falsification pattern.

Specifically, the shape of the metal nanoparticles may be sphere, hemisphere, ellipsoid, cube, cuboid, octahedron, dodecahedron, hexahedron, bar, star, cone, triangle, or cylinder.

A particle size of the metal nanoparticles may be in a range from 2 nm to 1 μm.

In some embodiments, the metal nanoparticles of the metal nanoparticles layer **120** are spaced apart from each other, and a gap of two adjacent metal nanoparticles is in a range from 2 nm to 1 mm, such that the metal nanoparticles are provides with a certain free space, thus the LSPR is generated the scattered light enhanced by the metal nanoparticles layer **120** under certain conditions.

In some embodiments, the filtering film layer **110** may define a Fabry Perot interference cavity.

Further, in this embodiment, the metal nanoparticles layer **120** is arranged on main surfaces of a front and rear sides of the filtering film layer **110**, such that the magnetic pigment flake **100** is arranged with a symmetrical structure centering on the filter film layer **110**.

A plurality of embodiments based on the first embodiment will be described in details as follows. Specifically, the filter film layer **110** of the magnetic pigment flake **100** will be described in details. It should be noted that the following embodiments can be combined arbitrarily without departing from technical ideas of the present disclosure.

As shown in FIG. **2**, FIG. **2** is a structural view of a magnetic pigment flake according to a second embodiment of the present disclosure.

A magnetic pigment flake **200** includes a filtering film layer **210** having the magnetic or magnetizable material and a metal nanoparticles layer **220**. The metal nanoparticles layer **220** is formed on a surface of the filtering film layer **210** and configured to generate the scattered light enhanced by the local surface plasmon resonance under an irradiation of light exceeding a predetermined intensity, that is, the local surface plasmon resonance is equal to the LSPR.

In some embodiments, the filtering film layer **210** includes a magnetic core layer **211** and a first medium layer **212**, and the first medium layer **212** is stacked on at least one of main surfaces of the magnetic core layer **211**. The metal nanoparticles layer **220** is arranged on a side of the first medium layer **212** away from the magnetic core layer **211**. An interference cavity is formed by the magnetic core layer **211** and the first medium layer **212**, so as to scatter, absorb, and/or diffuse the light, such that the light is re-scattered to the metal nanoparticles layer **220**, thus the scattered light enhanced by the LSPR is generated by the metal nanoparticles layer **220** under certain conditions.

In some embodiments, the magnetic core layer **211** has a magnetic or magnetizable material, such that the magnetic pigment flake **200** may be magnetically oriented under an effect of a magnetic field. A physical thickness of the magnetic core layer **211** is in a range from 2 nm to 10000 nm. In some embodiments, a range of the physical thickness of the magnetic core layer **211** is greater than 30 nm, such as 38 nm, 53 nm, 80 nm, or the like.

In some embodiments, the magnetic or magnetizable material of the magnetic core layer **211** are selected from at least one of iron, cobalt, nickel, gadolinium, terbium, dysprosium, erbium, alloys and oxides thereof. Alternatively, the magnetized material is selected from iron-silicon alloy, iron-aluminum alloy, iron-silicon-aluminum alloy, iron-silicon-chromium alloy, and iron-nickel-molybdenum alloy.

In some embodiments, the magnetic core layer **211** may be a single-layer structure.

In another embodiment, the magnetic core layer **211** may be a multi-layer composite structure, such as a M1M0M2 structure, a M0M1M0 structure, a M0M1 structure, a M1D1M0D2M2 structure, a M0D1M1D2M0 structure, a M1D1M0 structure, or the like. In some embodiment, the M0 is a magnetic film layer, and materials of the magnetic film layer are selected from iron, cobalt, nickel, gadolinium, terbium, dysprosium, erbium, alloys and oxides thereof. Alternatively, the materials of the magnetic film layer are selected from iron-silicon alloy, iron-aluminum alloy, iron-silicon-aluminum core, iron-silicon-chromium alloy, and iron-nickel-molybdenum, and alloys thereof. The M1 or M2 is a metal film layer, and materials of the metal film layer are selected from aluminum, silver, gold, copper, platinum, tin, titanium, palladium, rhodium, niobium, chromium, and alloys thereof. The D1 or D2 is a dielectric film layer, and the materials of the dielectric film layer are selected from silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, and calcium fluoride. By changing a variable structure of the magnetic core layer **211**, an interference effect of the interference cavity on the light can be more changeable, such that the anti-falsification pattern made by the magnetic pigment flake **200** can be more changeable, thereby improving the anti-falsification and imitation difficulty of the anti-falsification pattern.

In addition, a refraction index of the first medium layer **212** is greater than a threshold, such that light intensity of the light re-scattered into the metal nanoparticles layer **220** by the filtering film layer **210**, thus a color change generated by the metal nanoparticles layer **220** under the LSPR may be detected by the human eyes, thereby improving the anti-falsification the anti-falsification pattern. In some embodiments, the threshold may be set based on specific situations, such as 1.65 or 1.80.

Further, the first medium layer **212** may be made of at least one of lanthanum titanate, titanium pentoxide, niobium pentoxide, zinc sulfide, zinc oxide, zirconia, titanium dioxide, carbon, indium oxide, indium tin oxide, tantalum pentoxide, cerium oxide, yttrium oxide, europium oxide, iron oxide, ferroferric oxide, hafnium nitride, hafnium carbide, hafnium oxide, lanthanum oxide, magnesium oxide, neodymium oxide, praseodymium oxide, samarium oxide, antimony trioxide, silicon carbide, silicon nitride, silicon monoxide, selenium trioxide, tin oxide, and tungsten trioxide.

In some embodiments, a physical thickness of the first medium layer **212** is in a range from 30 nm to 80 nm. The physical thickness may be 20 nm, 25 nm, 50.5 nm, 70 nm, 85 nm, 99 nm, 100 nm, or the like.

In some embodiments, a coefficient of the film thicknesses of the first medium layer **212** may be less than or equal to 6, such as 0, 2, 4, or the like.

As shown in FIG. **3**, in this embodiment, the first medium layers **212** are arranged on main surfaces of a front and rear sides of the magnetic core layer **211** of the magnetic pigment flake **200**. Further, the metal nanoparticles layer **220** is arranged on a side surface of each of the first medium layers **212** away from the magnetic core layer **211**, such that the magnetic pigment flake **200** is arranged with a five-layer symmetrical structure centering on the magnetic core layer **211**. In some embodiments, as shown in FIG. **3**, two first medium layers **212** of the magnetic pigment flake **200** may have the same materials or different materials.

As shown in FIG. **4**, FIG. **4** is a structural view of a magnetic pigment flake according to a third embodiment of the present disclosure.

The magnetic pigment flake **300** includes a filtering film layer **310** having a magnetic or magnetizable material and a metal nanoparticles layer **320**. The metal nanoparticles layer **320** is formed on a surface of the filtering film layer **310** and configured to generate the scattered light enhanced by the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity.

In some embodiments, the filtering film layer **310** includes a magnetic core layer **311** and a first medium layer **312**, and the first medium layer **312** is stacked on at least one of main surfaces of the magnetic core layer **311**. In some embodiments, the metal nanoparticles layer **320** is arranged on a side of the first medium layer **312** away from the magnetic core layer **311**. Properties and structures of the magnetic core layer **311** and the first medium layer **312** may refer to the second embodiment, which will not be repeated here.

The filtering film layer **310** further includes a second medium layer **313**. The second medium layer **313** is arranged between the first medium layer **312** and the magnetic core layer **311**. In addition, a refraction index of the second medium layer **313** is less than that of the first medium layer **312**.

In the embodiments, a stacking structure formed by different medium layers with different refraction index is arranged on at least one of the main surfaces of the magnetic core layer **311**, such that a multi-color composite development effect is achieved, thus the magnetic pigment flake **300** has a higher reflectance and lower valley, so as to reduce an effect of the ink on the color development and an effect of an interface inversion, thereby further increasing a color development effect of the magnetic pigment flake **300** in the ink. In this way, a reflectance to ultraviolet rays of the magnetic pigment flake **300** is also increased, such that the color change generated by LSPR is more obvious when the irradiation light is ultraviolet light, thereby increasing an anti-aging performance of the magnetic pigment flake **300**, and reducing costs.

In some embodiments, the refraction index of the second medium layer **313** is less than a threshold. In some embodiments, the threshold may be set based on the specific situations, such as 1.65 or 1.80.

In some embodiments, materials of the second medium layer **313** include at least one of silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, calcium fluoride, lithium fluoride, polystyrene, polyethylene, polymethyl methacrylate (PMMA), polyamide imide, polyperfluoroethylene propylene, tetrafluoroethylene, trifluorochloroethylene, cellulose propionate, cellulose acetate, cellulose acetate butyrate, methylpentene polymer, homoformaldehyde, acrylic resin, cellulose nitrate, ethyl cellulose, polypropylene, polysulfone, poly-ethersulfone, mica, heterogeneous isomorphic polymer, polybutene, ionic cross-linked polymer, acrylic copolymer, thermoplastic, styrene butadiene, polyvinyl chloride (PVC), urea formaldehyde, styrene acrylonitrile, and polycarbonate.

In some embodiments, a physical thickness of the second medium layer **313** is greater than 30 nm. Specifically, the physical thickness is in a range from 80 nm to 180 nm, such as 80 nm, 100 nm, 105 nm, 150.5 nm, 160 nm, 180 nm, or the like.

In some embodiments, a coefficient of the film thicknesses of the second medium layer **313** may be less than or equal to 6, such as 0, 2, 4, or the like.

Further, a first dielectric stack may be arranged on at least one of the main surfaces of the magnetic core layer **311**, and the first dielectric stack is formed by a plurality of the first medium layer **312** and the second medium layer **313**. In this way, a plurality of film layers are stacked, such that the multi-color composite development effect is achieved, and at the same time, a translucent color development effect is also achieved. The first medium layer **312** disposed on the main surfaces of the magnetic core layer **311** and the second medium layer **313** are arranged alternately. In each first dielectric stack, the second medium layer **313** is closer to the magnetic core layer **311** than the first medium layer **312**, thereby meeting design requirements of a film system.

In addition, in this embodiment, the first dielectric stack may include at least three medium layers with different refractive indices, so as to improve uniformity of the refractive index change of the first dielectric stack. For example, as shown in FIG. **16**, the first dielectric stack may further include a third medium layer **314** arranged between the first medium layer **312** and the second medium layer **313**. A refractive index of the third medium layer **314** is less than that of the first medium layer **312**, and greater than that of the second medium layer **313**.

As shown in FIG. **5**, in this embodiment, the first medium layers **312** are arranged on the main surfaces of a front and rear sides of the magnetic core layer **311**. Further, the metal nanoparticles layer **320** is arranged on a side surface of each of the first medium layers **312** away from the magnetic core layer **311**, and the second medium layer **313** is arranged between the each of the first medium layers **312** and the magnetic core layer **311**, such that the magnetic pigment flake **300** is arranged with a seven-layer symmetrical structure centering on the magnetic core layer **311**.

In some embodiments, a plurality of the second medium layers **313** of the magnetic pigment flake **300** may be made of the same materials or different materials.

As shown in FIG. **6**, FIG. **6** is a structural view of a magnetic pigment flake according to a fourth embodiment of the present disclosure.

The magnetic pigment flake **400** includes a filtering film layer **410** having a magnetic or magnetizable material and a metal nanoparticles layer **420**. The metal nanoparticles layer **420** is formed on a surface of the filtering film layer **410** and configured to generate the scattered light enhanced by the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity.

In some embodiments, the filtering film layer **410** includes a magnetic core layer **411** and a first medium layer **412**, and the first medium layer **412** is stacked on at least one of main surfaces of the magnetic core layer **411**. In some embodiments, the metal nanoparticles layer **420** is arranged on a side of the first medium layer **412** away from the magnetic core layer **411**. Properties and structures of the magnetic core layer **411** and the first medium layer **412** may refer to the second embodiment, which will not be repeated here.

The filtering film layer **410** further includes an absorbing layer **414** and a second medium layer **413**. The absorbing layer **414** and the second medium layer **413** are arranged between the magnetic core layer **411** and the first medium layer **412**. The absorbing layer **414** is closer to the first medium layer **412** than the second medium layer **413**.

An interference cavity is formed by the magnetic core layer **411**, the second medium layer **413**, the absorbing layer **414**, and the first medium layer **412**, so as to scatter, absorb, and/or diffuse the light, such that the light is re-scattered to the metal nanoparticles layer **420**, thus the scattered light enhanced by the LSPR is generated by the metal nanoparticles layer **420** under certain conditions. Light in the specific wavelength band is absorbed by the absorbing layer **414**, such that the spectrum becomes narrower, thereby increasing a color saturation and color development chromaticity of the magnetic pigment flake **400**. Further, a magnetic reflective layer may be protected by the second medium layer **413**, so as to improve whole tolerance of the magnetic pigment flake **400**.

The absorbing layer **414** may have a translucent property, such that a part of the light may pass through the absorbing layer **414**. A thickness of the absorbing layer **414** is limited to make the absorbing layer **414** meet the translucent property. Specifically, a physical thickness of the absorbing layer **414** is less than 30 nm, such as 3 nm, 5 nm, 8 nm, 10 nm, 12 nm, 15 nm, 18 nm, or 20 nm.

Materials of the absorbing layer **414** may be selected from at least one of titanium, aluminum, chromium, nickel, palladium, titanium, vanadium, cobalt, iron, carbon, tin, tungsten, molybdenum, rhodium, niobium, and silicon carbide. In other words, the absorbing layer **414** may be made of the above elemental material, or alloy material composed of the above elements.

In addition, a refractive index of the second medium layer **413** is not limited, and may be greater than or equal to a threshold. Alternatively, the refractive index may be less than the threshold. The threshold may be set based on the specific situations, such as 1.65 or 1.80, which is not limited herein.

Materials of the second medium layer **413** may be at least one of silicon dioxide, magnesium fluoride, titanium dioxide, aluminium oxide, silicon monoxide, and cryolite.

In some embodiments, the second medium layer **413** may be a dielectric film layer, or a dielectric stack with alternating high and low refractive index film layers.

In some embodiments, a coefficient of the film thicknesses of the second medium layer **413** may be less than or equal to 6, such as 0, 2, 4, or the like.

In addition, a second dielectric stack may be arranged between the first medium layer **412** and the magnetic core layer **411**, and the second dielectric stack is formed by a plurality of the second medium layer **413** and the absorbing layer **414**. In this way, a plurality of film layers are stacked, such that the multi-color composite development effect is achieved, and at the same time, the translucent color development effect is also achieved. In some embodiments, the magnetic core layer **411**, the second dielectric stack arranged on the at least one of main surfaces of the magnetic core layer **411**, the first medium layer **412**, and the metal nanoparticles basically determine a main color and a final color of the magnetic pigment flake **400**, and the main color is viewing in a front view, and the final color is viewing at a viewing angel deviating from a normal direction of the main surface of the magnetic core layer **411** by 90 degrees or close to 90 degrees. Based on this case, an additional interference cavity may be formed by the additional second dielectric stack, such that a reflection peak in the specific wavelength band is selectively absorbed and filtered. In this way, when the viewing angel changes, a half width of the reflection peak in the specific wavelength band becomes narrower, such that an obvious color boundary is formed between different colors that can be observed, which is reflected in the color as discrete changes.

The reason why a color of the magnetic pigment flake **400** shows discrete changes is that due to a selective absorption of the absorbing layer **414** and the second medium layer **413** back and forth for many times, it plays a filtering role and filters out stray light near light with the highest reflection wavelength, such that when an interference occurs, light waves of a specific wavelength are constructively interfered, and light of wavelengths near the specific wavelength and other wavelengths are largely suppressed due to being filtered. In this way, when the viewing angel changes, a half width of the reflection peak in the specific wavelength band becomes narrower, such that the obvious color boundary is formed between the different colors that can be observed, which is reflected in the color as discrete changes.

The magnetic pigment flake **400** of the present disclosure has at least four colors. For example, when the viewing angel deviates from the normal direction of the main surface of the magnetic core layer **411** by 0 degree or close to 0 degree, for example, in a range from 0 degree to 5 degrees, the magnetic pigment flake **400** has a first color. When the viewing angel deviates from the normal direction of the main surface of the magnetic core layer **411** greater than 0 degree or 5 degrees and less than or equal to 45 degrees, the magnetic pigment flake **400** has a second color. When the viewing angel deviates from the normal direction of the main surface of the magnetic core layer **411** greater than 45 degrees and less than or equal to 90 degrees, the magnetic pigment flake **400** has a third color. The obvious color boundary is formed between the first color, the second color, and the third color. When the viewing angel is any degrees, a fourth color may be generated by the dark area formed by the magnetic pigment flake **400** and the LSPR of the metal nanoparticles layer **420**, and the fourth color is the hidden color. In this way, in cooperation with the metal nanoparticles layer **420**, a richer colors change is formed by the magnetic pigment flake **400**, thereby improving the anti-falsification and imitation difficulty of the anti-falsification pattern formed by the magnetic pigment flake **400**.

In some embodiments, in each second dielectric stack, the second medium layer **413** is closer to the magnetic core layer **411** than the absorbing layer **414**.

As shown in FIG. **7**, in this embodiment, the first medium layers **412** are arranged on the main surfaces of a front and rear sides of the magnetic core layer **411**. Further, the metal nanoparticles layer **420** is arranged on a side surface of each of the first medium layers **412** away from the magnetic core layer **411**, and the absorbing layer **414** and the second medium layer **413** are arranged between the each of the first medium layers **412** and the magnetic core layer **411**, such that the magnetic pigment flake **400** is arranged with a nine-layer symmetrical structure centering on the magnetic core layer **411**.

As shown in FIG. **8**, FIG. **8** is a structural view of a magnetic pigment flake according to a fifth embodiment of the present disclosure.

The magnetic pigment flake **500** includes a filtering film layer **510** having a magnetic or magnetizable material and a metal nanoparticles layer **520**. The metal nanoparticles layer **520** is formed on a surface of the filtering film layer **510** and configured to generate the scattered light enhanced by the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity.

In some embodiments, the filtering film layer **510** includes a first medium layer **512**, a magnetic core layer **511**, and a second medium layer **512** sequentially stacked. The metal nanoparticles layer **520** is arranged on a side of the first medium layer **512** away from the magnetic core layer **511**.

An interference cavity is formed by the second medium layer **513**, the magnetic core layer **511**, and the first medium layer **512**, so as to scatter, absorb, and/or diffuse the light, such that the light is re-scattered to the metal nanoparticles layer **520**, thus the LSPR is generated by the metal nanoparticles layer **520** under certain conditions. Further, in this embodiment, the filtering film layer **510** is configured to reflect the second medium layer **513**. In this way, it is not necessary to increase a thickness of the magnetic core layer **511** in order to improve a reflectance of the filter film layer **510**. Further, the filter film layer **510** is thinner, such that it is not easily to form a demagnetizing field inside the filter film layer **510**, thus there are fewer magnetic domain, energy of a magnetic domain wall is higher, and a magnetic permeability is increased, so as to make the magnetic pigment flake **500** easier to magnetize and have a better color development effect.

In some embodiments, properties and structures of the first medium layer **512** may refer to the second embodiment, which will not be repeated here.

In some embodiments, the magnetic core layer **511** has a magnetic or magnetizable material, such that the magnetic pigment flake **500** may be magnetically oriented under an effect of a magnetic field. A physical thickness of the magnetic core layer **511** is in a range from 2 nm to 10000 nm. In some embodiments, a range of the physical thickness of the magnetic core layer **511** is less than 30 nm, such as 2 nm, 10 nm, 15 nm, 20 nm, or the like, such that at least a part of the light may pass through the magnetic core layer **511**. Further, a thickness of the magnetic core layer **511** is thinner, such that it is not easily to form the demagnetizing field inside the filter film layer **510**, thus there are fewer magnetic domain, the energy of the magnetic domain wall is higher, and the magnetic permeability is increased, so as to make the magnetic pigment flake **500** easier to magnetize and have the better color development effect.

In some embodiments, the magnetic core layer **511** may be a single-layer structure. Alternatively, the magnetic core layer **511** may be a multi-layer composite structure.

In addition, a refractive index of the second medium layer **513** is not limited, and may be greater than or equal to a threshold. Alternatively, the refractive index may be the threshold. The threshold may be set based on the specific situations, such as 1.65 or 1.80, which is not limited herein.

Materials of the second medium layer **513** may be at least one of silicon dioxide, magnesium fluoride, titanium dioxide, aluminium oxide, silicon monoxide, and cryolite.

In some embodiments, the second medium layer **513** may be the dielectric film layer, or the dielectric stack with alternating high and low refractive index film layers.

Further, as shown in FIG. **9**, in this embodiment, the number of the magnetic core layer **511** is at least two. The second medium layer **513** is arranged between two adjacent magnetic core layers **511**. The first medium layer **512** is arranged on a main surface of the outermost magnetic core layer **511** away from the second medium layer **513**. The metal nanoparticles layer **520** is arranged on a main surface of the first medium layer **512** away from the second medium layer **513**, such that the magnetic pigment flake **500** is arranged with a seven-layer symmetrical structure centering on the second medium layer **513**. In this case, magnetic moments between at least two magnetic core layers **511** affect each other, such that when the magnetic pigment sheet **500** is magnetized, a better color development effect is formed.

As shown in FIG. **10**, FIG. **10** is a structural view of a magnetic pigment flake according to a sixth embodiment of the present disclosure.

The magnetic pigment flake **600** includes a filtering film layer **610** having a magnetic or magnetizable material and a metal nanoparticles layer **620**. The metal nanoparticles layer **620** is formed on a surface of the filtering film layer **610** and configured to generate the scattered light enhanced by the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity.

In some embodiments, the filtering film layer **610** includes a first medium layer **612**, a magnetic core layer **611**, a second medium layer **613**, and a reflective layer **614** sequentially stacked. The metal nanoparticles layer **620** is arranged on a side of the first medium layer **612** away from the magnetic core layer **611**.

An interference cavity is formed by the second medium layer **613**, the magnetic core layer **611**, the first medium layer **612**, and the reflective layer **614**, so as to scatter, absorb, and/or diffuse the light, such that the light is re-scattered to the metal nanoparticles layer **620**, thus the LSPR is generated by the metal nanoparticles layer **620** under certain conditions. Further, in this embodiment, the filtering film layer **610** is configured to reflect the reflective layer **614**. In this way, it is not necessary to increase a thickness of the magnetic core layer **611** in order to improve a reflectance of the filter film layer **610**. Further, the filter film layer **610** is thinner, such that it is not easily to form a demagnetizing field inside the filter film layer **610**, thus there are fewer magnetic domain, the energy of the magnetic domain wall is higher, and the magnetic permeability is increased, so as to make the magnetic pigment flake **600** easier to magnetize and have the better color development effect.

In some embodiments, properties and structures of the first medium layer **612** may refer to the second embodiment, which will not be repeated here. Properties and structures of the second medium layer **613** may refer to the fifth embodiment, which will not be repeated here.

In some embodiments, the magnetic core layer **611** has a magnetic or magnetizable material, such that the magnetic pigment flake **600** may be magnetically oriented under an effect of a magnetic field. A physical thickness of the magnetic core layer **611** is in a range from 2 nm to 10000 nm. In some embodiments, a range of the physical thickness of the magnetic core layer **611** is less than 30 nm, such as 2 nm, 10 nm, 16 nm, 20 nm, or the like, such that at least a part of the light may pass through the magnetic core layer **611**. Further, a thickness of the magnetic core layer **611** is thinner, such that it is not easily to form the demagnetizing field inside the filter film layer **610**, thus there are fewer magnetic domain, the energy of the magnetic domain wall is higher, and the magnetic permeability is increased, so as to make the magnetic pigment flake **600** easier to magnetize and have the better color development effect.

In some embodiments, the magnetic core layer **611** may be a single-layer structure. Alternatively, the magnetic core layer **611** may be a multi-layer composite structure.

In addition, the reflective layer **614** may be made of metal. Further, materials of the reflective layer **614** may be selected from aluminum, silver, gold, copper, platinum, tin, titanium, palladium, rhodium, niobium, chromium, and alloys thereof.

A physical thickness of the reflective layer **614** is in a range from 2 nm to 500 nm, such as 30 nm, 50 nm, or 80 nm.

Further, as shown in FIG. **11**, in this embodiment, the number of the magnetic core layer **611** is at least two. The second medium layer **613** and the reflective layer **614** are arranged between two adjacent magnetic core layers **611**. Further, the reflective layer **614** is separated from each of the two adjacent magnetic core layers **611** by the second medium layer **613**. The first medium layer **612** is arranged on a main surface the outermost magnetic core layer **611** away from the second medium layer **613**. The metal nanoparticles layer **620** is arranged on a main surface of the first medium layer **612** away from the second medium layer **613**, such that the magnetic pigment flake **600** is arranged with a nine-layer symmetrical structure centering on the reflective layer **614**. In this case, magnetic moments between at least two magnetic core layers **611** affect each other, such that when the magnetic pigment sheet **600** is magnetized, a better color development effect is formed.

In the plurality of embodiments as described above, the magnetic pigment flake may be manufactured by a process, such as physical vapor deposition, chemical vapor deposition, sol-gel method, dipping method, and so on.

In some embodiments, in the plurality of the embodiments as described above, the magnetic pigment flake may be mixed with an ink body, so as to manufacture an optically variable ink.

Of course, the optically variable ink of the magnetic pigment flake of the present disclosure may be coated on an article body to manufacture an anti-falsification article. The magnetic pigment flake of the optically variable ink is magnetically oriented, such that the bright and dark areas are generated with the viewing angel changing under an irradiation of light below the predetermined intensity. Further, under the irradiation of the visible light exceeding the predetermined intensity, light with a color different from that of the bright area is generated on a corresponding position of the dark area.

In order to describe an effect of the metal nanoparticles layer of the present disclosure, a following comparative example and a plurality of examples are provided by the present disclosure.

A Comparative Example

A base layer is provided. Specifically, the base layer may be a rigidity or flexible base. For example, materials of the base layer may be a quartz glass or polyethylene terephthalate (PET). An isolation layer is formed on the base layer. A first medium layer, a second medium layer, a metal film layer, a magnetic film layer, the metal film layer, the second medium layer, and the first medium layer are sequentially deposited on the isolation layer. Specifically, on the rigidity base, the isolation layer, the first medium layer, the second medium layer, the metal film layer, the magnetic film layer, the metal film layer, the second medium layer, and the first medium layer are set to cycles. In this case, vapor deposition is repeatedly performed for 20 to 30 times, or more times. The magnetic pigment flake deposited on the base layer is detached by a dry process or wet process, the magnetic pigment flake after grinded is mixed with the ink so as to manufacture the optically variable ink, and the optically variable ink is printed on a substrate. A pattern of the optically variable ink printed on the substrate is magnetized. It may be observed that an effect of the pattern of the optically variable ink after magnetization. As shown in FIG. **12**, by observing the effect of the pattern of the optically variable ink made in the comparative example after magnetization, it may be seen that when the optically variable ink pattern is rotated under the ambient light, the dark area of the pattern is gradually displaced, the pattern has a rolling effect, and a pattern color gradually changes from blue to purple.

A First Example

A base layer is provided. Specifically, the base layer may be a rigidity or flexible base. For example, materials of the base layer may be the quartz glass or PET. An isolation layer is formed on the base layer. A metal nanoparticles layer, a first medium layer, a second medium layer, a metal film layer, a magnetic film layer, the metal film layer, the second medium layer, the first medium layer, and the metal nanoparticles layer are sequentially deposited on the isolation layer. Specifically, on the rigidity base, the isolation layer, the metal nanoparticles layer, the first medium layer, the second medium layer, the metal film layer, the magnetic film layer, the metal film layer, the second medium layer, the first medium layer, and the metal nanoparticles layer are set to cycles. In this case, vapor deposition is repeatedly performed for 20 to 30 times, or more times. The magnetic pigment flake deposited on the base layer is detached by the dry process or wet process, the magnetic pigment flake after grinded is mixed with the ink so as to manufacture the optically variable ink, the optically variable ink is printed on a substrate. A pattern of the optically variable ink printed on the substrate is magnetized. It may be observed that an effect of the pattern of the optically variable ink after magnetization. As shown in FIG. **13**, it may be observed that an effect of the pattern of the optically variable ink after magnetization. As shown in FIG. **14**, it may be observed that a reflectance-wavelength spectrum of the optically variable ink after magnetization. Besides, an ordinate or y-axis incidents the reflectance, and an abscissa or x-axis incidents the wavelength. A thin line indicates a spectral curve of the optically variable ink pattern in the front view direction, for example, the viewing angel deviating from the normal direction of the main surface of the magnetic core layer by 5 degrees. A thick line indicates a spectral curve of the optically variable ink pattern in a side view direction, for example, the viewing angel deviating from the normal direction of the main surface of the magnetic core layer by 60 degrees. In this case, it may be seen that when the optically variable ink pattern is rotated under the ambient light, the dark area of the pattern is gradually displaced, and the pattern has a rolling effect. At the same time, under the ambient light, when the viewing angle changes from 5 degrees to 60 degrees, a pattern color gradually changes from blue to purple. Further, combined with a scattering spectrum of the optically variable ink after scraping under strong light shown in FIG. **15** and an observation in FIG. **13**, it can be seen that under an irradiation of strong visible light, enhanced scattered light is generated in the patterned dark area due to the LSPR of the metal nanoparticles, such that a color of the dark area of the pattern becomes yellow, while a color of the bright area remains unchanged. When the pattern is rotated, the blue, namely, the color of the bright area, and the yellow color, namely, the color of the dark area, are gradually displaced, thereby generating the rolling effect, which may be changed alternately. Therefore, under an irradiation of the strong visible light, there is no obvious dark area, and the color of the pattern is composed of the color of the bright area and the hidden color.

A Second Example

A base layer is provided. Specifically, the base layer may be a rigidity or flexible base. For example, materials of the base layer may be the quartz glass or PET. An isolation layer is formed on the base layer. A metal nanoparticles layer, a first medium layer, a second medium layer, a metal film layer, a magnetic film layer, the metal film layer, the second medium layer, the first medium layer, and the metal nanoparticles layer are sequentially deposited on the isolation layer. Specifically, on the rigidity base, the isolation layer, the metal nanoparticles layer, the first medium layer, the second medium layer, the metal film layer, the magnetic film layer, the metal film layer, the second medium layer, the first medium layer, and the metal nanoparticles layer are set to a circle. In this case, vapor deposition is repeatedly performed for 20 to 30 times, or more times. The magnetic pigment flake deposited on the base layer is detached by the dry process or wet process, so as to obtain the magnetic pigment flake.

A Third Example

By using a winding coating device, a base layer is provided. Specifically, the base layer may be a flexible base. For example, materials of the base layer may be the PET. An isolation layer is formed on the base layer. A metal nanoparticles layer, a first medium layer, a second medium layer, a metal film layer, a magnetic film layer, the metal film layer, the second medium layer, the first medium layer, and the metal nanoparticles layer are sequentially deposited on the isolation layer. A specific detaching method may be the dry process, the wet process, a transfer printing, or a method of using an adhesive base to attach and detach.

A Fourth Example

By using the winding coating device, a base layer is provided. Specifically, the base layer may be a flexible base. For example, materials of the base layer may be the PET. An isolation layer is formed on the base layer. A first medium layer, an absorbing layer, a second medium layer, a magnetic film layer, the second medium layer, the absorbing layer, and the first medium layer are sequentially deposited on the isolation layer. The specific detaching method may be the dry process, the wet process, the transfer printing, or the method of using the adhesive base to attach and detach. In this case, metal nanoparticles grow up on a pigment by a chemical process.

The above are only embodiments of the present disclosure and are not intended to limit the scope of the present disclosure. Any equivalent structural changes made under the concept of the present disclosure, using the contents of the specification of the present disclosure and the accompanying drawings, or applied directly/indirectly in other related fields of technology are included in the scope of protection of the present disclosure.

What is claimed is:

1. A magnetic pigment flake, comprising:

a filtering film layer, with a magnetic or magnetizable material; and

a metal nanoparticles layer, formed on a surface of the filtering film layer;

wherein the metal nanoparticles layer has a light scatter-enhancing property with a wavelength selectivity for incident light, the metal nanoparticles layer is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity, and the filtering film layer has a Fabry Perot interference cavity; a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light exceeding the predetermined intensity is different from a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light below the predetermined intensity.

2. The magnetic pigment flake according to claim **1**, wherein metal nanoparticles layer has a scattered color in response to being irradiated by the visible light being greater than 3000 lx and has a reflection color in response to being irradiated by the visible light being lower than 300 lx.

3. The magnetic pigment flake according to claim **1**, wherein a particle size of the metal nanoparticles is in a range from 2 nm to 1 μm;

materials of the metal nanoparticles are selected from at least one of aluminum, silver, gold, copper, platinum, ruthenium, palladium, rhodium, cobalt, iron, nickel, lead, osmium, iridium, and alloys thereof;

a shape of the metal nanoparticles is sphere, hemisphere, ellipsoid, cube, cuboid, octahedron, dodecahedron, hexahedron, bar, star, cone, triangle, or cylinder.

4. The magnetic pigment flake according to claim **1**, wherein the filtering film layer comprises a magnetic core layer and a first medium layer stacked on the magnetic core layer, the metal nanoparticles layer is arranged on a side surface of the first medium layer away from the magnetic core layer, a refraction index of the first medium layer is greater than 1.65; and

the metal nanoparticles of the metal nanoparticles layer are spaced apart from each other, and a gap of two adjacent metal nanoparticles is in a range from 2 nm to 1 mm.

5. The magnetic pigment flake according to claim **1**, wherein the metal nanoparticles layer is arranged on each of a front surface and a rear side of the filtering film layer, such that the magnetic pigment flake is arranged with a symmetrical structure centering on the filtering film layer.

6. The magnetic pigment flake according to claim **1**, wherein the filtering film layer comprises a magnetic core layer, a first medium layer, and a second medium layer; the first medium layer is stacked on at least one of main surfaces of the magnetic core layer, the metal nanoparticles layer is arranged on a side of the first medium layer away from the magnetic core layer, the second medium layer is arranged between the first medium layer and the magnetic core layer, and a refraction index of the second medium layer is less than a refraction index of the first medium layer.

7. The magnetic pigment flake according to claim **6**, wherein the magnetic core layer is a multi-layer composite structure, and the multi-layer composite structure comprises at least one of a M1M0M2 structure, a M0M1M0 structure, a M1D1M0D2M2 structure, a M0D1M1D2M0 structure, and a M1D1M0 structure, the M0 is a magnetic film layer, the M1 or M2 is a metal film layer, and the D1 or D2 is a dielectric film layer.

8. The magnetic pigment flake according to claim **7**, wherein the refraction index of the second medium layer is less than or equal to 1.65;

the second medium layer is selected from at least one of silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, calcium fluoride, lithium fluoride, polystyrene, polyethylene, polymethyl methacrylate, polyamide imide, polyperfluoroethylene propylene, tetrafluoroethylene, trifluorochloroethylene, cellulose propionate, cellulose acetate, cellulose acetate butyrate, methylpentene polymer, homoformaldehyde, acrylic resin, cellulose nitrate, ethyl cellulose, polypropylene, poly-sulfone, poly-ethersulfone, mica, heterogeneous isomorphic polymer, polybutene, ionic cross-linked polymer, acrylic copolymer, thermoplastic, styrene butadiene, polyvinyl chloride, urea formaldehyde, styrene acrylonitrile, and polycarbonate.

9. The magnetic pigment flake according to claim **7**, wherein the filtering film layer further comprises an absorbing layer; the absorbing layer and the second medium layer are arranged between the magnetic core layer and the first medium layer; the absorbing layer is closer to the first medium layer than the second medium layer.

10. The magnetic pigment flake according to claim **9**, wherein a thickness of the absorbing layer is less than 30 nm;

materials of the absorbing layer are selected from at least one of titanium, aluminum, chromium, nickel, palladium, titanium, vanadium, cobalt, iron, carbon, tin, tungsten, molybdenum, rhodium, and niobium, or selected from an alloy of at least one thereof, or selected from silicon carbide.

11. The magnetic pigment flake according to claim **7**, wherein-the number of the magnetic core layer is at least two; the second medium layer is arranged between two adjacent magnetic core layers; the first medium layer is arranged on a main surface of the outermost magnetic core layer away from the second medium layer.

12. The magnetic pigment flake according to claim **11**, wherein a physical thickness of the magnetic core layer is less than 30 nm;

the second medium layer is a dielectric stack with alternating high and low refractive index film layers, or a dielectric film layer;

materials of the second medium layer are selected from at least one of silicon dioxide, aluminium oxide, magnesium fluoride, aluminum fluoride, cerium fluoride, lanthanum fluoride, neodymium fluoride, samarium fluoride, barium fluoride, and calcium fluoride.

13. The magnetic pigment flake according to claim **7**, wherein the filtering film layer further comprises a reflective layer; the number of the magnetic core layer is at least two; the second medium layer and the reflective layer are arranged between two adjacent magnetic core layers; the reflective layer is separated from each of the two adjacent magnetic core layers by the second medium layer; the first medium layer is arranged on a main surface of the outermost magnetic core layer away from the second medium layer.

14. The magnetic pigment flake according to claim **13**, wherein materials of the reflective layer are selected from at least one of aluminum, silver, gold, copper, platinum, tin, titanium, palladium, rhodium, niobium, chromium, and alloys thereof;

a physical thickness of the reflective layer is in a range from 2 nm to 500 nm.

15. The magnetic pigment flake according to claim **7**, wherein an interference cavity is formed by the magnetic core layer and the first medium layer.

16. The magnetic pigment flake according to claim **7**, wherein the first medium layer is selected from at least one of lanthanum titanate, titanium pentoxide, niobium pentoxide, zinc sulfide, zinc oxide, zirconia, titanium dioxide, carbon, indium oxide, indium tin oxide, tantalum pentoxide, cerium oxide, yttrium oxide, europium oxide, iron oxide, ferroferric oxide, hafnium nitride, hafnium carbide, hafnium oxide, lanthanum oxide, magnesium oxide, neodymium oxide, praseodymium oxide, samarium oxide, antimony trioxide, silicon carbide, silicon nitride, silicon monoxide, selenium trioxide, tin oxide, and tungsten trioxide;

the magnetic core layer is made of at least one of iron, cobalt, nickel, gadolinium, terbium, dysprosium and erbium, oxides and alloys thereof.

17. An optically variable ink, comprising:

an ink body; and

magnetic pigment flakes, each of the magnetic pigment flakes comprising:

a filtering film layer, with a magnetic or magnetizable material; and

a metal nanoparticles layer, formed on a surface of the filtering film layer;

wherein each of metal nanoparticles layer has a light scatter-enhancing property with a wavelength selectivity for incident light, the each of the metal nanoparticles layers is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity, and the filtering film layer has a Fabry Perot interference cavity; a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light exceeding the predetermined intensity is different from a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light below the predetermined intensity and

wherein the magnetic pigment flakes are doped in the ink body.

18. An anti-falsification article, comprising:

an article body; and

an optically variable ink, coated on the article body, and comprising an ink body and magnetic pigment flakes;

wherein each of the magnetic pigment flakes comprises:

a filtering film layer, with a magnetic or magnetizable material; and

a metal nanoparticles layer, formed on a surface of the filtering film layer;

wherein each of metal nanoparticles layer has a light scatter-enhancing property with a wavelength selectivity for incident light, the each of the metal nanoparticles layers is configured to generate scattered light enhanced by a local surface plasmon resonance under an irradiation of visible light exceeding a predetermined intensity, and the filtering film layer has a Fabry Perot interference cavity; a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light exceeding the predetermined intensity is different from a color of the metal nanoparticles layer under the local surface plasmon resonance in response to the metal nanoparticles layer being irradiated by the visible light below the predetermined intensity and

wherein the magnetic pigment flake of the optically variable ink is magnetically oriented, such that a bright and dark areas are generated with a viewing angel changing under an irradiation of visible light below the predetermined intensity; and under an irradiation of visible light exceeding the predetermined intensity, light with a color different from that of the bright area is generated on a corresponding position of the dark area.

19. The anti-falsification article according to claim **18**, wherein the anti-falsification article is configured to generate scattered light enhanced by the local surface plasmon resonance under an irradiation of the light exceeding a predetermined intensity, such that a color generated by the dark area of the anti-falsification article under an irradiation of the light exceeding a predetermined intensity is different from a color generated by the dark area of the anti-falsification article under an irradiation of the light below the predetermined intensity.

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